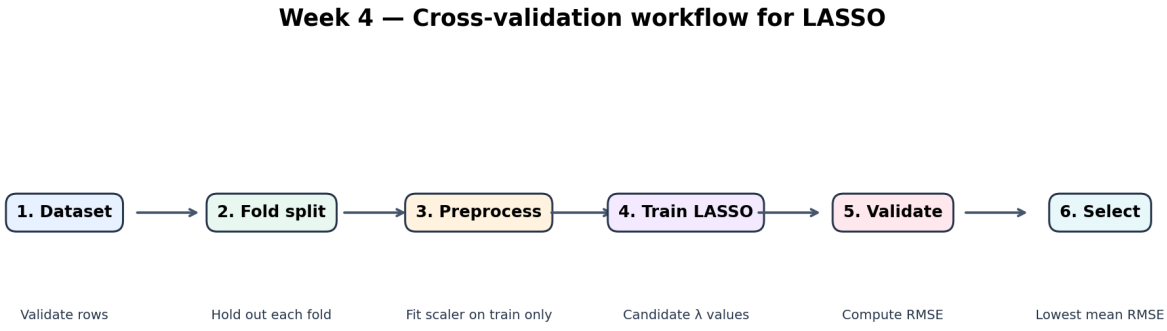
LASSO REGRESSION

WEEK 4: VALIDATION & MODEL IMPROVEMENT

C++17 project extension • Technical report

Focus: deterministic k-fold cross-validation, λ selection using validation RMSE, and leakage-aware preprocessing.



Educational project report. Included data are synthetic and are not commercial telemetry or proprietary company data.

1. Problem and objectives

A model that performs well on training data may generalise poorly. Week 4 compares candidate regularisation parameters using held-out folds instead of relying only on training cost. Objectives: implement reproducible k-fold validation, avoid validation leakage, select λ by mean RMSE, and preserve earlier project modules.

2. Method

For each candidate λ, the data rows are shuffled using a fixed seed and partitioned into k folds. Each fold becomes validation data once. The scaler is fitted only on the remaining training rows, then applied to both training and validation features. Coordinate descent trains the model, and validation RMSE is recorded. The selected λ minimises mean fold RMSE.

3. Metric

RMSE = sqrt((1/n) Σ(y■ – ■■)²). RMSE is expressed in the target variable's units and penalises larger errors more heavily than MAE. It should be interpreted alongside data quality, residual behaviour and an untouched final test set.

4. New implementation

`include/lasso/week4.hpp` defines `CrossValidationResult` and the public API. `src/member10_cross_validation.cpp` implements fold generation, fold-wise scaling, model fitting and λ selection. `examples/week4_demo.cpp` runs an illustrative energy-data example. `tests/test_week4.cpp` checks deterministic output, valid scores, parameter selection and invalid fold counts.

5. OOP and design

The project retains the introductory `LassoModel` class and modular functions for scaling, fitting, prediction and evaluation. Week 4 uses a result struct to group related values and a small API to encapsulate validation mechanics. This demonstrates modularity and data abstraction; it does not falsely claim inheritance or runtime polymorphism.

6. Engineering and technology context

A future electrical-engineering dataset might relate ambient temperature, distance, payload, battery current or energy consumption. Cross-validation helps evaluate generalisation, but the example here uses synthetic values only. Tesla AI/robotics is included as context for why data-driven engineering is relevant—not as a claim that Tesla uses this student code. This is not a vehicle, robot, battery or industrial controller.

7. Results and limitations

Run the demo to print the selected λ, each fold's RMSE, mean validation RMSE and training-fit diagnostics. The dataset is small and synthetic, so the numbers are not evidence of commercial performance. Training metrics shown after λ selection are not independent test metrics. Future work should use an untouched test set, repeated or nested cross-validation, residual plots, and a larger public dataset.

8. Build and test

Run: `cmake -S . -B build`, then `cmake --build build`, then `ctest --test-dir build --output-on-failure`. Finally run `./build/lasso_week4_demo` (Windows: `build\\lasso_week4_demo.exe`). The report's Markdown counterpart is `week-04.md`.